Embedded Systems Programming

College of Science and Technology

School Of ICT

Computer Science Department

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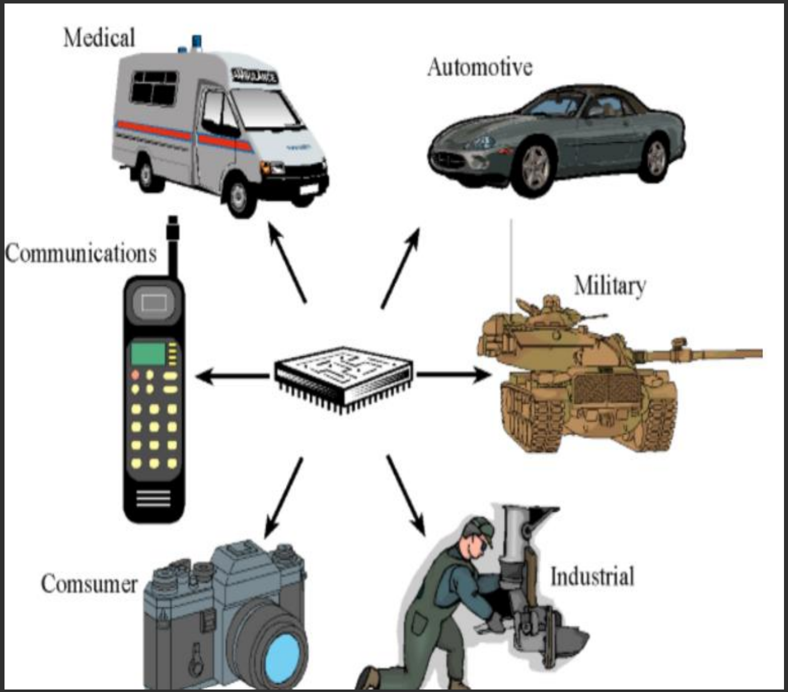
Overview

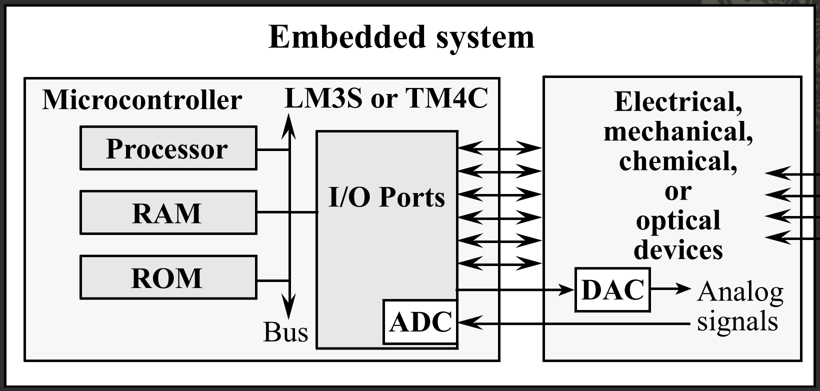
- What is an embedded system(ES)?
  - An embedded system is a microcomputer with mechanical, chemical, and electrical devices attached to it, programmed for a specific dedicated purpose, and packaged up as a complete system
    - Or
  - "An embedded system is an application that contains at least one programmable computer (typically in the form of a microcontroller, a microprocessor or digital signal processor chip) and which is used by individuals who are, in the main, unaware that the system is computer-based." Michael J. Pont, Embedded C
- Embedded Systems are everywhere
  - Ubiquitous, invisible
  - Hidden (computer inside)
  - Dedicated purpose

- A combination of computer hardware and software, and perhaps additional mechanical or other parts, designed to perform a dedicated function

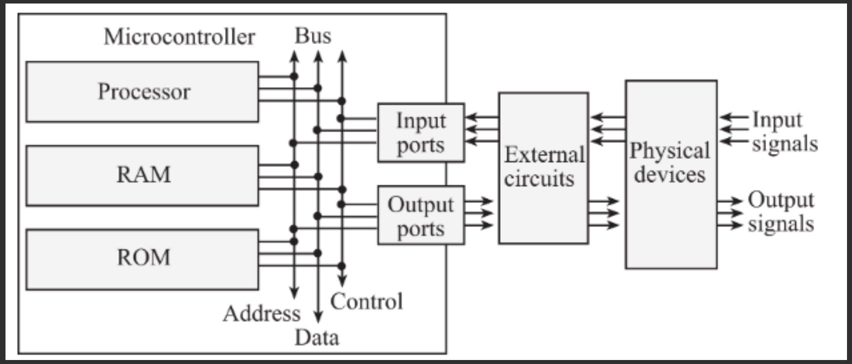


Source

- **Embedded systems** on the other hand are optimized for a single function or a narrow range of tasks



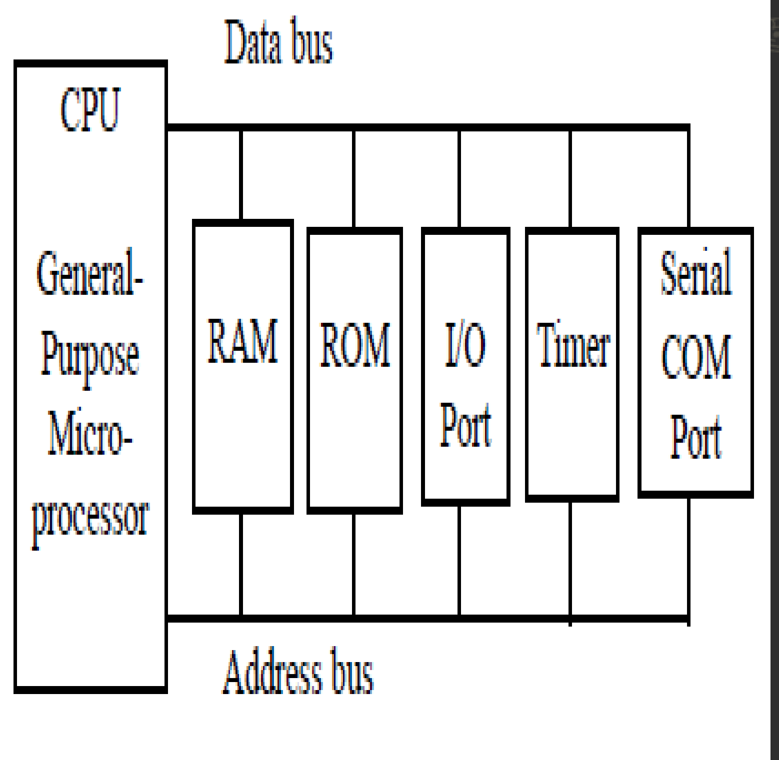
- An embedded system uses a **microprocessor** (or **microcontroller**) to do **one task** and one task only



An embedded system includes a microcomputer interfaced to external physical devices

Microprocessor

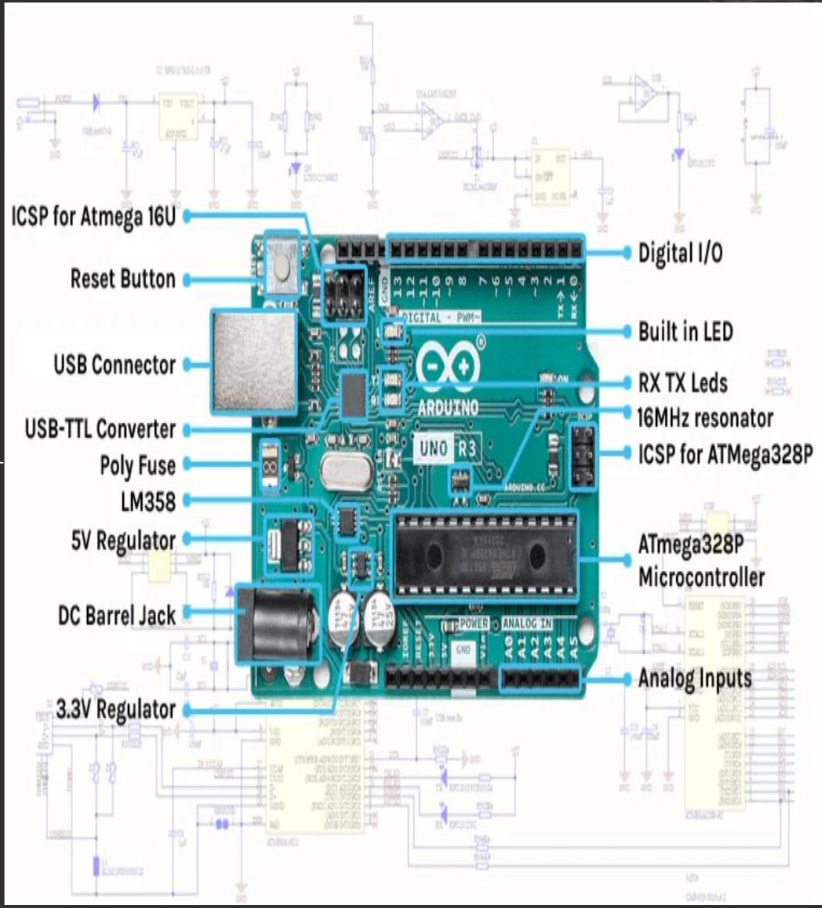
- For this reason, they are commonly referred to as general-purpose microprocessors.
- A system designer using a microprocessor must add (**interface**) RAM, ROM, I/O ports, and timers externally to make it functional.



(a) General-Purpose Microprocessor System

What is a microcontroller?

- "A microcontroller is a computer-on-a-chip, or, if you prefer, a single-chip computer. *Micro* suggests that the device is small, and *controller* tells you the device might be used to control objects, processes, or events. Another term to describe a microcontroller is *embedded controller*, because the microcontroller and its support circuits are often built into, or embedded in, the devices they control."
The Microcontroller Idea Book, Jan Axelson



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ES Application

- Consumer electronics: Washing machine, Exercise equipment, Remote controls, Clocks and watches, Games and toys, Audio/video electronics, Set-back thermostats, Camera, Camcorder, Television, VCR, cable box

Remote controls

PlayStation

Washing machine

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Application…

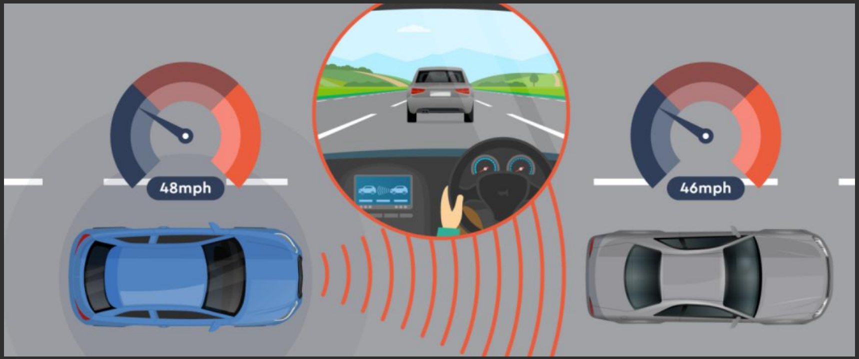
- <u>Communication systems</u>: Answering machines, Telephones, Fax machines, Radios, Cellular phones, pagers

[Source]

Application …..

- <u>Automotive systems</u>: Automatic breaking, Noise cancellation, Locks, Electronic ignition, Power windows and seats, Cruise control, Collision avoidance, Climate control, Emission control,

Elevator lift



Cruise control

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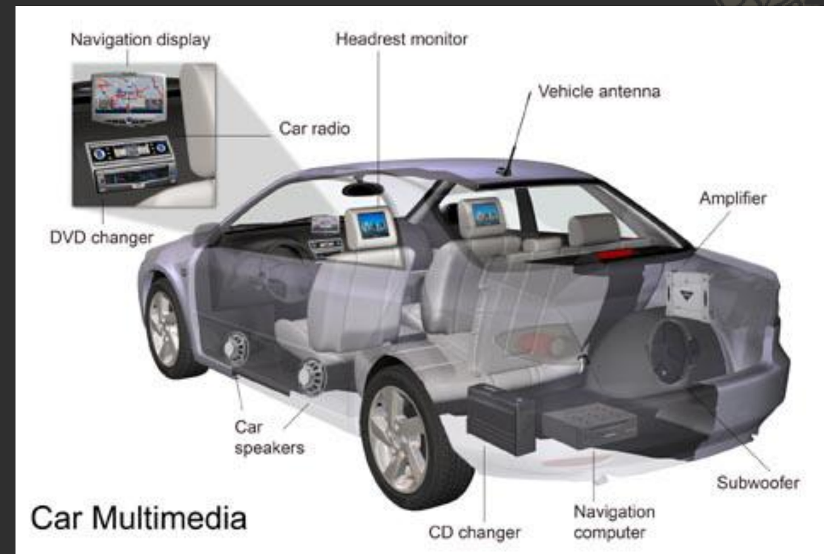
Application..

- Transportation



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- Military hardware:
  - Smart weapons,
  - Missile guidance systems,
  - Global positioning systems,
  - Surveillance systems

Smart weapon

Missile guidance systems

Unmanned aerial vehicle(UAV)

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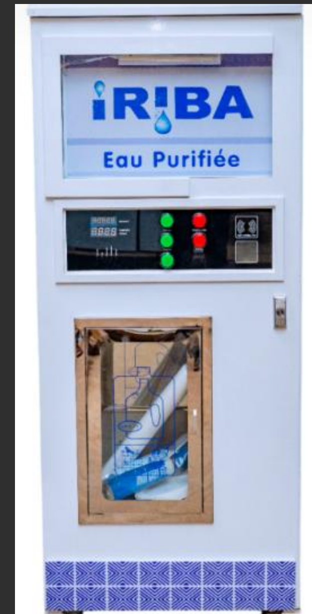
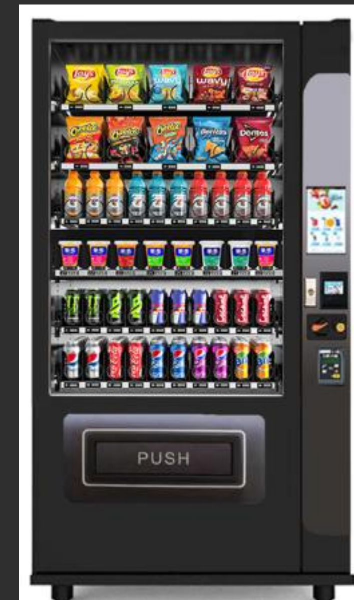
- Business applications: Cash registers, Vending machines, ATM machines, Traffic controllers, Industrial robots, Bar code readers and writers, Automatic sprinklers, Elevator controllers, RFID systems, Lighting and heating systems
- Medical devices: Monitors, Drug delivery systems, Cancer treatments, Pacemakers, Prosthetic devices, Dialysis machines

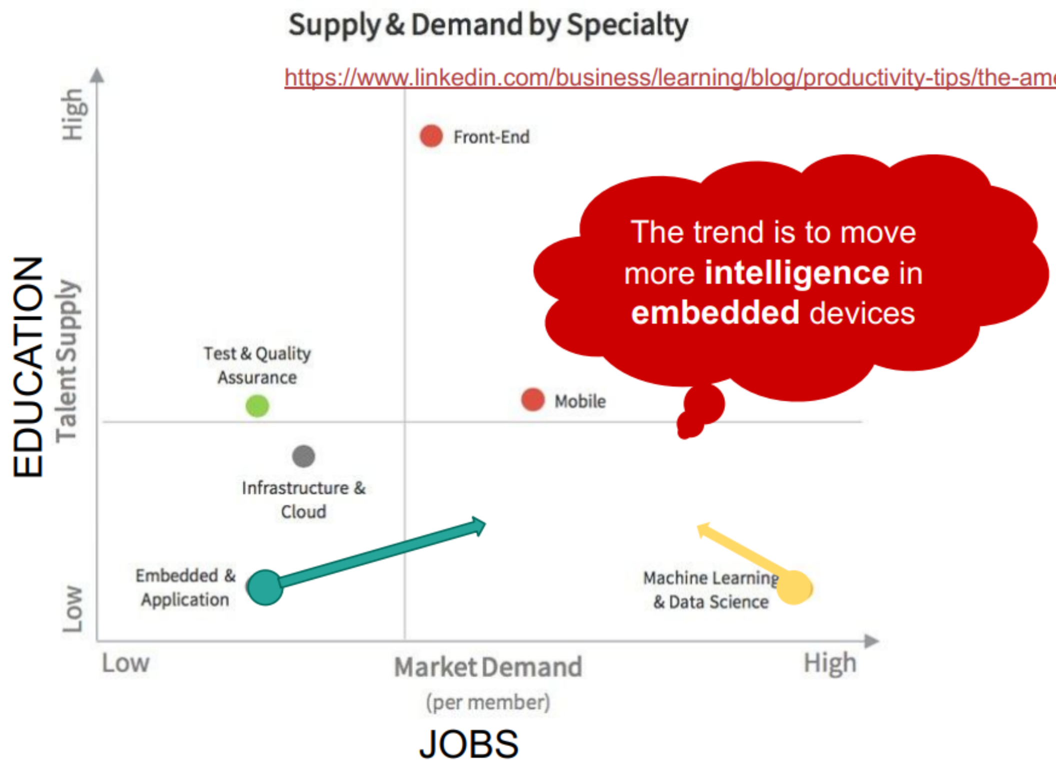


Iriba



Vending machine

Opportunity E S



Software Engineering Speciality	% of engineers in 2018
Frontend	38
Mobile	18
Test & Quality Assurance	18
Infrastructure & Cloud	15
Embedded & Application	**5**
Machine Learning & Data Science	5

Only 5% of engineers have embedded skills (according to Linkedin - 2024), while 95% of software are embedded ...

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General Computer System vs ES

- General Purpose Computing System
  - A system which is a combination of generic hardware and General Purpose Operating System for executing a variety of applications
  - Contain a General Purpose Operating System (GPOS)
- ESA
  - System which is a combination of special purpose hardware and embedded OS for executing a specific set of applications
  - May or may not contain an operating system for functioning

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General Com.. System vs ES……..

- General Purpose Computing System
  - Applications are alterable (programmable) by user (It is possible for the end user to re-install the Operating System, and add or remove user applications)
  - Performance is the key deciding factor on the selection of the system. Always 'Faster is Better'
  - Less/not at all tailored towards reduced operating power requirements, options for different levels of power management

- ES
  - The firmware of the embedded system is pre-programmed and it is non-alterable by end-user
  - Application specific requirements (like performance, power requirements, memory usage etc) are the key deciding factors
  - Highly tailored to take advantage of the power saving modes supported by hardware and Operating System

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General Com.. System vs ES……..

- General Purpose Computing System
  - Response requirements are not time critical
  - Need not be deterministic in execution behavior

- ES
  - For certain category of embedded systems like mission critical systems, the response time requirement is highly critical
  - Execution behavior is deterministic for certain type of embedded systems like 'Hard Real Time' systems

History of Embedded System

- Embedded System (ESs) existed even before IT revolution.
  - In the olden days, ESs were built using vacuum tubes and transistor technology.
  - Embedded algorithms were developed in low level languages
- The first modern, real-time ES was “Apollo Guidance Computer (AGC)”. It was developed in 1960 by Dr. Charles Stark Draper at the MIT Instrumentation Labs

AGC

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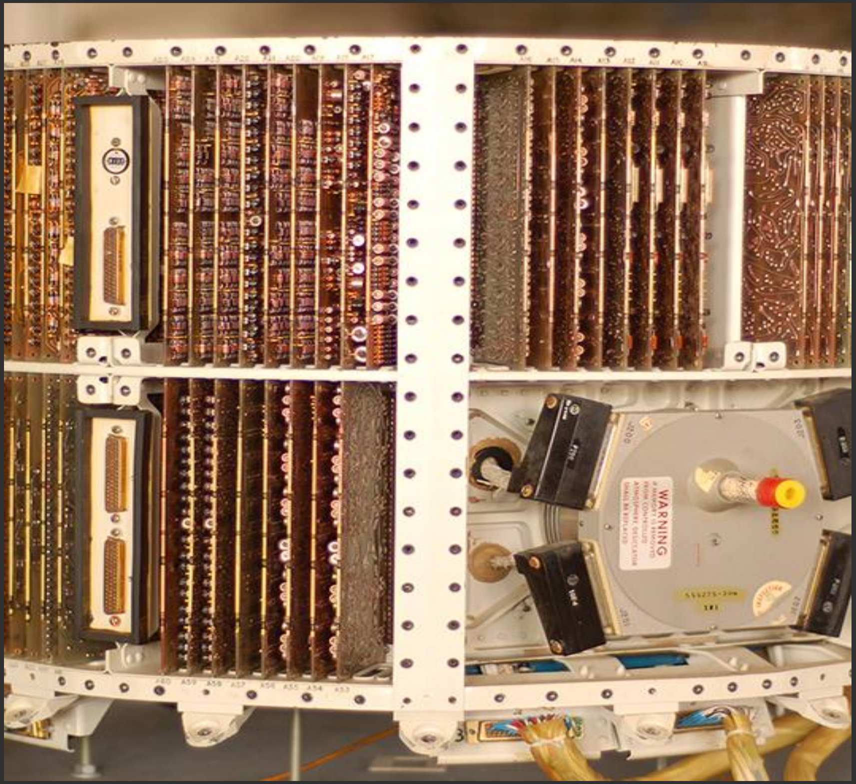
History of Embedded System….

- In 1965, Autonetics developed the D-17B.
  - It was used in the computer used in the Minuteman missile guidance system.
  - It is widely recognized as the first mass produced ES



[Autonetics D-17B Guidance Computer]

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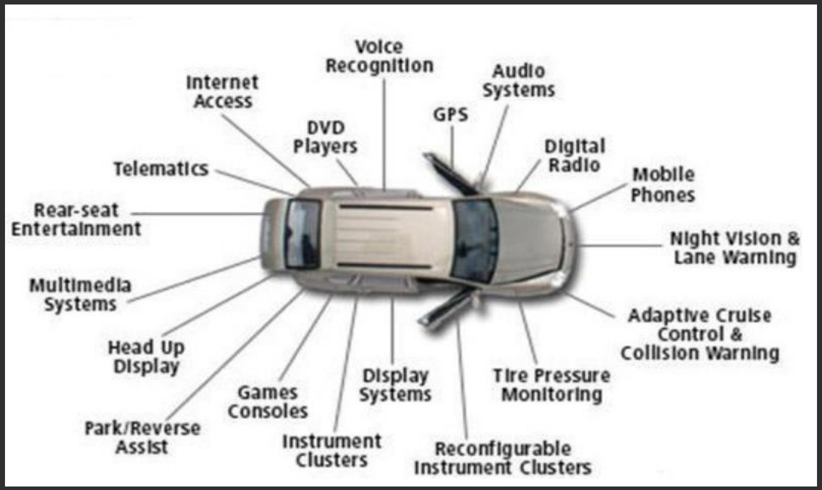
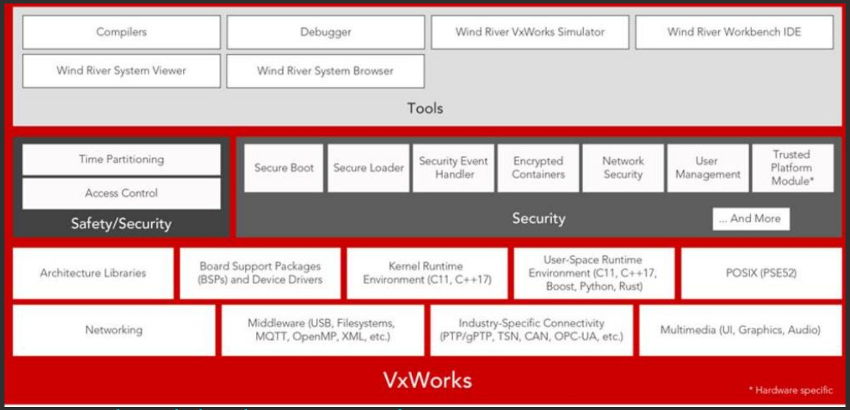
History of Embedded System….

- In 1968, the first ES for a vehicle was released.
- In 1971, Texas Instruments developed the first microcontroller.
- In 1987, the first embedded OS, VxWorks, was released by Wind River.
- In 1996, Microsoft developed its OS called “Windows embedded Compact”.
- By the late 1990s, the first embedded Linux system appeared



Embedded System in Automobile



Embedded Operating System

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Es Classification

1. ES based on Generation
2. Classification based on Complexity & Performance
3. Classification Based on deterministic behavior:
4. **Classification Based on Triggering:**

Classification of ES

1. ES based on Generation

- This classification is based on the order in which the embedded processing systems evolved from the first version to where they are today
  - **First Generation:**
    - The early embedded systems were built around 8bit microprocessors like 8085 and Z80, and 4bit microcontrollers.
    - Simple in hardware circuits with firmware developed in Assembly code.
    - Example:
      - Digital telephone keypads, stepper motor control units etc.

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Classification of ES

1. ES based on Generation …

- **Second Generation**
  - These are embedded systems built around 16bit microprocessors and 8 or16 bit microcontrollers, following the first generation embedded systems.
  - The instruction set for the second generation processors/controllers were much more complex and powerful than the first generation processors/ controllers.
  - Some of the second generation embedded systems contained embedded operating systems for their operation.
  - Examples : Data Acquisition Systems and SCADA(Supervisory Control and Data Acquisition systems) used in industry plant, etc. are second generation embedded systems.

Classification of ES

1. ES based on Generation ....

- **Third Generation**

With advances in processor technology, embedded system developers started making use of powerful 32bit processors and 16bit microcontrollers for their design.

- Introduction of domain specific processors/controllers like Digital Signal Processors (DSP) and Application Specific Integrated Circuits (ASICs) came into the picture
- The instruction set of processors became more complex and powerful and the concept of instruction pipelining also evolved.
- The processor market was flooded with different types of processors from different vendors. Processors like Intel Pentium, Motorola 68K, etc. gained attention in high performance embedded requirements.
- Dedicated embedded real time and general-purpose operating systems entered into the embedded market.
- Embedded systems spread its ground to areas like robotics, media, industrial process control, networking, etc

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Classification of ES

1. ES based on Generation ....

- **Fourth Generation**
  - System on Chips (SoC), reconfigurable processors and multicore processors started to be used in ES and are brought high performance, tight integration and miniaturisation into the embedded device market.
  - The SoC(System On Chip) technique implements a total system on a chip by integrating different functionalities with a processor core on an integrated circuit.
  - The fourth generation embedded systems are making use of high-performance real time embedded operating systems for their functioning.
  - Example
    - Smart phone devices, mobile internet devices (MIDs), etc.
- **What Next Generation?**
  - The processor and embedded market are highly dynamic and demanding.
  - So many sophisticated applications will come in next generation
  - What will be the next smart move in the next embedded generation?' Let's wait and see

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Classification of ES

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  - The processor and embedded market are highly dynamic and demanding.
  - So many sophisticated applications will come in next generation

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Classification of ES…..

2. Classification based on Complexity & Performance

- Small Scale:
  - The embedded systems built around low performance and low cost 8 or 16 bit microprocessors/ microcontrollers. It is suitable for simple applications and where performance is not time critical. It may or may not contain OS.
- Medium Scale:
  - Embedded Systems built around medium performance, low cost 16 or 32 bit microprocessors / microcontrollers or DSPs. These are slightly complex in hardware and firmware. It may contain GPOS/RTOS.
- Large Scale/Complex:
  - Embedded Systems built around high performance 32 or 64 bit RISC processors/controllers, RSoC or multi-core processors and PLD. It requires complex hardware and software. These system may contain multiple processors/controllers and co-units/hardware accelerators for offloading the processing requirements from the main processor. It contains RTOS for scheduling, prioritizationand management.

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3. Classification Based on deterministic behavior:

- It is applicable for Real Time systems.
  - The application/task execution behavior for an embedded system can be either deterministic or non-deterministic
- Are Classified in two :
  - Soft embedded systems (or soft real-time systems) are computing systems with time-sensitive tasks where missing a deadline reduces service quality but does not cause system failure
  - Hard Real-Time Embedded Systems - In these types of embedded systems time/deadline of task is strictly followed. Task must be completed in between time frame (defined time interval) otherwise result/output may not be accepted.

4. Classification Based on Triggering:

- These are classified into two types
- **Event Triggered :** Activities within the system (e.g., task run-times) are dynamic and
  depend upon occurrence of different events .
- **Time triggered:** Activities within the system follow a statically computed schedule (i.e., they are allocated time slots during which they can take place) and thus by nature are predictable.

Assignment

- Group 1& 2 make slides for class1 of embedded system by indicating the application
- Group 3& 4 make slides for class2 of embedded system by indicating the application
- Group 5& 6 make slides for class3 of embedded system by indicating the application
- Group 7make slides for class4 of embedded system by indicating the application

Purpose of Embedded Systems:

- **Data Collection/Storage/Representation**
- **Data Communication**
- Data (Signal) Processing
- **Monitoring**
- Control
- **Application Specific User Interface**

Purpose of Embedded Systems:

- **Data Collection/Storage/Representation**
  - Performs acquisition of data from the external world
  - Performs acquisition of data from the external world.
  - Data collection is usually done for storage,analysis, manipulation and transmission
  - The collected data may be stored directly in the system or may be transmitted to some other systems or it may be processed by the system or it may be deleted instantly after giving a meaningful representation

Source

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Purpose of Embedded Systems:

- **Data Communication**
  - Embedded Data communication systems are deployed in applications ranging from complex satellite communication systems to simple home networking systems
  - Embedded Data communication systems are dedicated for data communication
  - The data communication can happen through a

 wired interface (like Ethernet, RS- 232C/USB/IEEE1394 etc) or wireless interface (like Wi-Fi, GSM,/GPRS, Bluetooth, ZigBee etc)

  - Network hubs, Routers, switches, Modems etc are typical examples for dedicated data transmission embedded systems

router

GSM/GPRS

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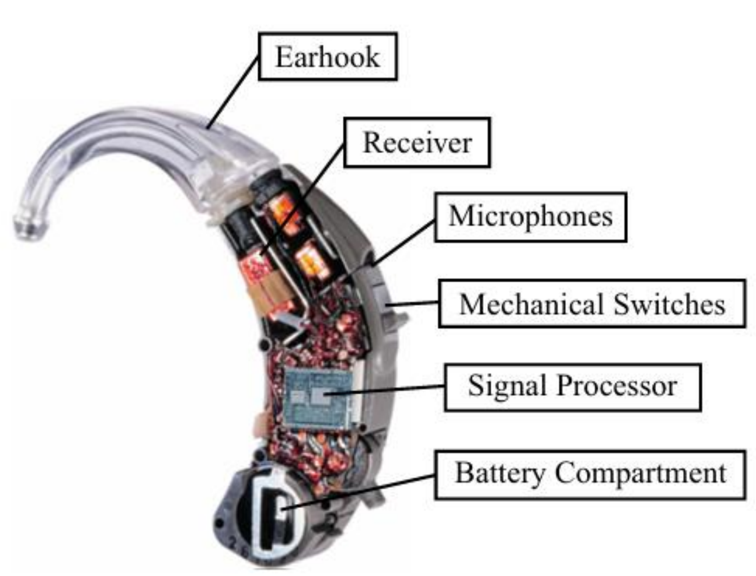
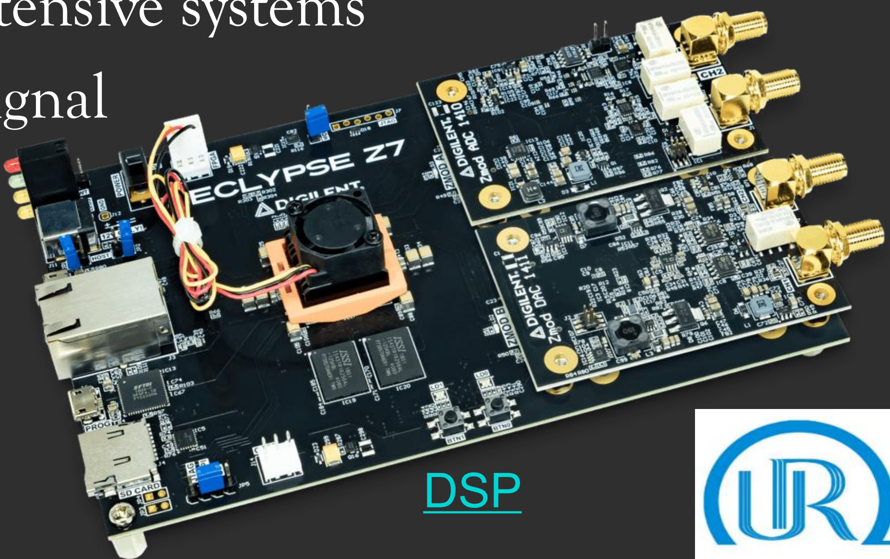
Purpose of Embedded Systems:

- Data (Signal) Processing
  - Embedded systems with Signal processing functionalities are employed in applications demanding signal processing like Speech coding, synthesis, audio video codec, transmission applications etc
  - Computational intensive systems
  - Employs Digital Signal Processors (DSPs)



Digital Hearing Aids



DSP

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Purpose of Embedded Systems:

- **Monitoring**
  - Embedded systems coming under this category are specifically designed for monitoring purpose
  - They are used for determining the state of some variables using input sensors
  - Electro Cardiogram (ECG) machine for monitoring the heart beat of a patient is a typical example for this
  - The sensors used in ECG are the different Electrodes connected to the patient’s body
  - Measuring instruments like Digital CRO(Cathode Ray

Oscilloscope), Digital Multi meter, Logic Analyzer etc used in Control & Instrumentation applications for knowing (monitoring) the status of some variables like current, voltage are also examples of embedded systems for monitoring purpose

ECG

Vital Signal Monitoring

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Control

- ES with Control Functionality are used for imposing control over some variables according to the changes in input variables
- Embedded systems with control functionalities are used for imposing control over some variables
  according to the changes in input variables
- Embedded system with control functionality contains both sensors and actuators
- Sensors are connected to the input port for capturing the changes in environmental variable or measuring variable

Example automatic AC

Control….

- The actuators connected to the output port are controlled according to the changes in input variable to put an impact on the controlling variable to bring the controlled variable to the specified range
- Air conditioner for controlling room temperature is a typical example for embedded system with ‘Control’ functionality
- Air conditioner contains a room temperature sensing element (sensor) which may be a thermistor and a handheld unit for setting up (feeding) the desired temperature
- The air compressor unit acts as the actuator.
  - The compressor is controlled according to the current room temperature and the desired temperature set by the end user.

Automatic AC

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Application Specific User Interface

- Embedded systems which are designed for a specific application
- Contains Application Specific User interface (rather than general standard UI) like buttons, switches, keypad, lights, bells, keyboard, Display units, etc
- Aimed at a specific target group of users
- Mobile handsets, Control units in industrial applications etc

[Phone]

['Smart' running shoes from Adidas]

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CHARACTERISTICS OF EMBEDDED SYSTEMS

- Each ES has a set of specific and unique characteristics. Some of the important characteristics of ESs are:
  1. Application and Domain specific
  2. Reactive and Real Time
  3. Operates in harsh environments
  4. Distributed
  5. Small size and weight
  6. Power Concerns

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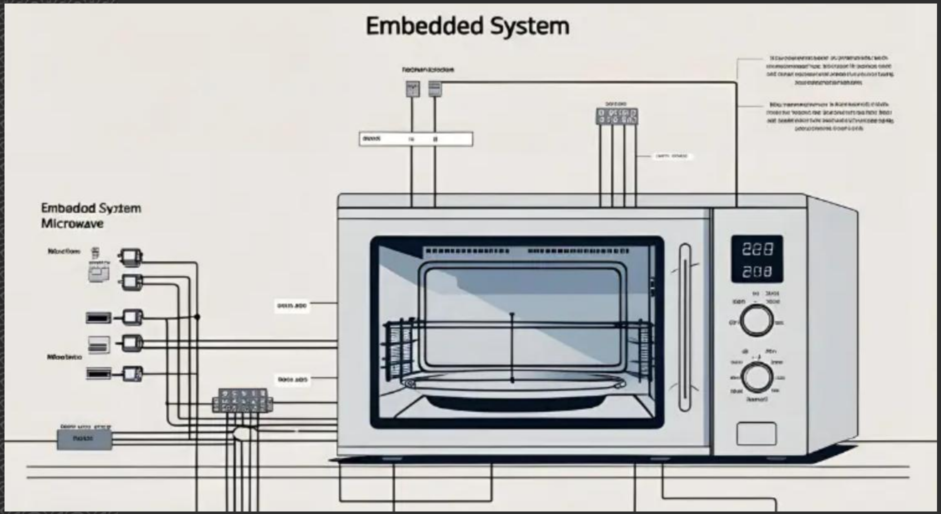
CHARACTERISTICS OF EMBEDDED SYSTEMS….

- characteristics of ESs are:
  1. Application and Domain specific
     - Each ES is designed to perform one or few unique functions. It cannot be used for any other purpose.
     - For example:
       - You cannot replace embedded control system
       - For a microwave oven by that of a washing machine.
       - For a Telecom domain with that of a consumer electronics domain.

Washing Machine



Microwave oven

CHARACTERISTICS OF EMBEDDED SYSTEMS…

2. Reactive and Real Time
   - ESs continuously interact with the Real world through sensors, control algorithms and input devices at the input port of the system.
   - They are also called reactive systems.
     - An ES produces changes in output response to the changes in the input variables.
     - It is designed with proper response time for each task.
     - It should not miss the deadline for any task.
     - Any change in the real world is called event.
   - An event may be a periodic event or unpredicted event.
     - There is no problem with periodic events.
     - They are well identified and handled.
     - The important thing is an ES should not miss unpredicted events.
     - The design of such systems should take worst-case situations into consideration.
   - Example of Real Time systems :flight control systems, Antilock Brake Systems (ABS), etc

Note: Some ESs may not be real time in operation. (e.g., electronic toys)

CHARACTERISTICS OF EMBEDDED SYSTEMS……

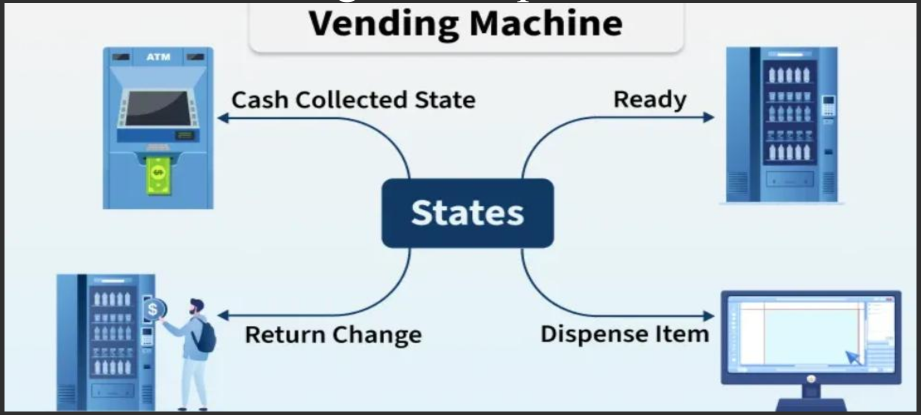
- characteristics of ESs are:

3. Operates in harsh environments
   - Sometimes ESs may be installed in harsh environments.
   - Such systems must be capable to withstand all the adverse conditions.
   - The design should take care of operating conditions of the area where it is installed.
     - Example 1: ESs in high temperature zone
       - All the components used in the system should be of high temperature grade.
       - Here we cannot go for compromise in cost.
     - Example 2: ESs in vibrations and shock environment
       - Proper shock absorption techniques should be provided.
     - Other examples for harsh environments: Power supply fluctuations, corrosion, component aging, etc.

CHARACTERISTICS OF EMBEDDED SYSTEMS………

- Characteristics of ESs :

4. Distributed
   - Embedded systems may be a part of larger systems
   - Distributed embedded systems form a single large embedded control unit
   - All of them are independent embedded units. But they work together to perform the specified function.
     - Example 1: Automatic chocolate vending machine (ACVM) contains :
       - a user keypad,
       - coin counting module,
       - and chocolate dispensing mechanism.
       - Each of them is an individual ES.
       - All of them work together to perform overall vending function.



VM

CHARACTERISTICS OF EMBEDDED SYSTEMS

5. Small size and weight
   - Product aesthetics is an important factor in choosing a product
     - example :
       - If You plan to buy a new mobile phone, you may make a comparative study on the pros and cons of the products available in the market
       - Definitely the product aesthetics (size, weight, shape, style, etc.) will be one of the deciding factors to choose a product
       - People believe in the phrase "Small is beautiful"
       - it is convenient to handle a compact device than a bulky product.
       - In embedded domain also compactness is a significant deciding factor.
       - Most of the application demands small sized and low weight products

CHARACTERISTICS OF EMBEDDED SYSTEMS

6. Power Concerns
   - Power consumption: Power consumption is the amount of energy used per unit time
     - This implies that ESs should have low power consumption
     - This can be achieved by using the ultra-low power consuming components available in the market
     - Also, battery life of portable systems (e.g., Cell phones and laptop computers) is limited by power consumption
     - Power management is another important factor that needs to be considered in designing embedded systems
   - Power dissipation:
     - An ES with large power dissipation needs cooling fans, which occupy additional space and make the system bulky.
     - ESs should have low power dissipation.
     - To achieve this, ESs are designed using regulators with low dropout and processors / controllers with power saving mode
     - Embedded systems should be designed in such a way as to minimise the heat dissipation by the system

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Further reading

- https://maitronics.com/introduction-to-embedded-systems/
- https://www.elprocus.com/basics-of-embedded-system-and-applications/
- https://intechhouse.com/blog/what-is-embedded-software-engineering/
- https://genxtechy.com/embedded-systems-history/
- https://www.theengineeringprojects.com/2021/05/types-of-embedded-systems.html#google_vignette
- https://www.ko2.co.uk/what-are-embedded-systems-used-for/
- https://www.advantech.com/pt-br/resources/industry-focus/embedded-system-essentials-definitions-and-uses
- Characteristics of Embedded Systems - The Engineering Projects
- What are the Practical Applications of Embedded Systems? - EE-Vibes
- Exploring Live Examples of Embedded Systems We Use in Daily Life – Yhills
- Taxonomy, technology and applications of smart objects | Information Systems Frontiers | Springer Nature Link

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